Further applications of the normal distribution

1

a 44% **b** 89% **c** 50 848

2

a 23% **b** 13% **c** 2524

3

a 0.769 **b** 70.060% **c** 31.757%

4

a 26.154 cm **b** 27.280 cm **c** 24.724 cm **d** 0.248

e 0.690

5

a 3.493 kg **b** 3.721 kg **c** 3.363 kg **d** 0.309

e 0.614

6

a 190.047 cm **b** 195.869 cm **c** 4.8% **d** 0.566

e 0.371

7

a $5.108, 5.012$

b 0.3

c There is a good chance that the machine is faulty.

8

a 301.15 K **b** 3.3 K

9

a 3500 g **b** 550 g

10

a 550 mm **b** 1600 mm

11

a $b = 4$ **b** 68.48

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12

a -328 **b** 16

13

a 2.985 **b** 67.848 g

14

a 34.457 cm **b** $(X+11)\text{ cm}$ **c** 34.457 cm

15

a $0.706=\dfrac{70-\mu}{\sigma}$ **b** $-0.151=\dfrac{56-\mu}{\sigma}$

c $\mu=58.465, \sigma=16.331$

16

a 58.7% **b** $(0.7\mu+2)\%$ **c** $(0.7\sigma)\%$

17

a 122.736 **b** $\dfrac{20}{37}X$

18

a 40% **b** 3 **c** 69%

19

a 40.1 **b** 0.69 **c** 0.035 **d** 0.346

e 1.74 mins

20

a 83% **b** 0.64 **c** 44%

21

a 0.43 **b** 172.83 g **c** 0.85 **d** 177.26 g

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